

## Lab Assignment 1: Data Wrangling

### Step 1: Import all the required Python Libraries.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os
```

### Step 2: Locate an open source data from the web.

#### Dataset Selected

#### Heart Disease Dataset

#### Source: Kaggle

Dataset: <https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>

```
import kagglehub
path = kagglehub.dataset_download("johnsmith88/heart-disease-dataset")
print("Dataset path:", path)
```

Using Colab cache for faster access to the 'heart-disease-dataset' dataset.  
Dataset path: /kaggle/input/heart-disease-dataset

### Step 3: Load the Dataset into pandas dataframe.

```
os.listdir(path)
```

```
['heart.csv']
```

```
df = pd.read_csv(os.path.join(path, "heart.csv"))
df.head()
```

|   | age | sex | cp | trestbps | chol | fbs | restecg | thalach | exang | oldpeak | slope | ca | thal | target |
|---|-----|-----|----|----------|------|-----|---------|---------|-------|---------|-------|----|------|--------|
| 0 | 52  | 1   | 0  | 125      | 212  | 0   | 1       | 168     | 0     | 1.0     | 2     | 2  | 3    | 0      |
| 1 | 53  | 1   | 0  | 140      | 203  | 1   | 0       | 155     | 1     | 3.1     | 0     | 0  | 3    | 0      |
| 2 | 70  | 1   | 0  | 145      | 174  | 0   | 1       | 125     | 1     | 2.6     | 0     | 0  | 3    | 0      |
| 3 | 61  | 1   | 0  | 148      | 203  | 0   | 1       | 161     | 0     | 0.0     | 2     | 1  | 3    | 0      |
| 4 | 62  | 0   | 0  | 138      | 294  | 1   | 1       | 106     | 0     | 1.9     | 1     | 3  | 2    | 0      |

### Step 4: Perform Data Preprocessing

```
df.shape
```

```
(1025, 14)
```

```
df.columns
```

```
Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach',  
      'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],  
      dtype='object')
```

```
df.dtypes
```

|          | 0       |
|----------|---------|
| age      | int64   |
| sex      | int64   |
| cp       | int64   |
| trestbps | int64   |
| chol     | int64   |
| fbs      | int64   |
| restecg  | int64   |
| thalach  | int64   |
| exang    | int64   |
| oldpeak  | float64 |
| slope    | int64   |
| ca       | int64   |
| thal     | int64   |
| target   | int64   |

dtype: object

df.describe()

|       | age         | sex         | cp          | trestbps    | chol        | fbs         | restecg     | thalach     | exang       | oldpeak     |
|-------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| count | 1025.000000 | 1025.000000 | 1025.000000 | 1025.000000 | 1025.000000 | 1025.000000 | 1025.000000 | 1025.000000 | 1025.000000 | 1025.000000 |
| mean  | 54.434146   | 0.695610    | 0.942439    | 131.611707  | 246.000000  | 0.149268    | 0.529756    | 149.114146  | 0.336585    | 1.071511    |
| std   | 9.072290    | 0.460373    | 1.029641    | 17.516718   | 51.59251    | 0.356527    | 0.527878    | 23.005724   | 0.472772    | 1.175051    |
| min   | 29.000000   | 0.000000    | 0.000000    | 94.000000   | 126.000000  | 0.000000    | 0.000000    | 71.000000   | 0.000000    | 0.000000    |
| 25%   | 48.000000   | 0.000000    | 0.000000    | 120.000000  | 211.000000  | 0.000000    | 0.000000    | 132.000000  | 0.000000    | 0.000000    |
| 50%   | 56.000000   | 1.000000    | 1.000000    | 130.000000  | 240.000000  | 0.000000    | 1.000000    | 152.000000  | 0.000000    | 0.800000    |
| 75%   | 61.000000   | 1.000000    | 2.000000    | 140.000000  | 275.000000  | 0.000000    | 1.000000    | 166.000000  | 1.000000    | 1.800000    |
| max   | 77.000000   | 1.000000    | 3.000000    | 200.000000  | 564.000000  | 1.000000    | 2.000000    | 202.000000  | 1.000000    | 6.200000    |

df.isnull().sum()

|          | 0 |
|----------|---|
| age      | 0 |
| sex      | 0 |
| cp       | 0 |
| trestbps | 0 |
| chol     | 0 |
| fbs      | 0 |
| restecg  | 0 |
| thalach  | 0 |
| exang    | 0 |
| oldpeak  | 0 |
| slope    | 0 |
| ca       | 0 |
| thal     | 0 |
| target   | 0 |

dtype: int64

```
df.fillna(df.median(numeric_only=True), inplace=True)
```

```
df.isnull().sum()
```

|          | 0 |
|----------|---|
| age      | 0 |
| sex      | 0 |
| cp       | 0 |
| trestbps | 0 |
| chol     | 0 |
| fbs      | 0 |
| restecg  | 0 |
| thalach  | 0 |
| exang    | 0 |
| oldpeak  | 0 |
| slope    | 0 |
| ca       | 0 |
| thal     | 0 |
| target   | 0 |

dtype: int64

### Step 5: Data Formatting and Data Normalization

```
categorical_cols = ['sex', 'cp', 'fbs', 'restecg',  
                   'exang', 'slope', 'ca', 'thal', 'target']
```

```
for col in categorical_cols:  
    df[col] = df[col].astype('category')
```

```
df.dtypes
```

|          | 0        |
|----------|----------|
| age      | int64    |
| sex      | category |
| cp       | category |
| trestbps | int64    |
| chol     | int64    |
| fbs      | category |
| restecg  | category |
| thalach  | int64    |
| exang    | category |
| oldpeak  | float64  |
| slope    | category |
| ca       | category |
| thal     | category |
| target   | category |

dtype: object

```
numeric_cols = ['age', 'trestbps', 'chol', 'thalach', 'oldpeak']
```

```
df[numeric_cols] = (df[numeric_cols] - df[numeric_cols].min()) / \  
    (df[numeric_cols].max() - df[numeric_cols].min())
```

```
(df[numerical_cols].max() - df[numerical_cols].min())
```

```
df.head()
```

|   | age      | sex | cp | trestbps | chol     | fbs | restecg | thalach  | exang | oldpeak  | slope | ca | thal | target |
|---|----------|-----|----|----------|----------|-----|---------|----------|-------|----------|-------|----|------|--------|
| 0 | 0.479167 | 1   | 0  | 0.292453 | 0.196347 | 0   | 1       | 0.740458 | 0     | 0.161290 | 2     | 2  | 3    | 0      |
| 1 | 0.500000 | 1   | 0  | 0.433962 | 0.175799 | 1   | 0       | 0.641221 | 1     | 0.500000 | 0     | 0  | 3    | 0      |
| 2 | 0.854167 | 1   | 0  | 0.481132 | 0.109589 | 0   | 1       | 0.412214 | 1     | 0.419355 | 0     | 0  | 3    | 0      |
| 3 | 0.666667 | 1   | 0  | 0.509434 | 0.175799 | 0   | 1       | 0.687023 | 0     | 0.000000 | 2     | 1  | 3    | 0      |
| 4 | 0.687500 | 0   | 0  | 0.415094 | 0.383562 | 1   | 1       | 0.267176 | 0     | 0.306452 | 1     | 3  | 2    | 0      |

#### Step 6: Turn categorical variables into quantitative variables in Python.

```
df_encoded = pd.get_dummies(df, columns=categorical_cols, drop_first=True)
```

```
df_encoded.head()
```

|   | age      | trestbps | chol     | thalach  | oldpeak  | sex_1 | cp_1  | cp_2  | cp_3  | fbs_1 | ... | slope_1 | slope_2 | ca_1  | ca_2  | ca_3  | ca_4  |
|---|----------|----------|----------|----------|----------|-------|-------|-------|-------|-------|-----|---------|---------|-------|-------|-------|-------|
| 0 | 0.479167 | 0.292453 | 0.196347 | 0.740458 | 0.161290 | True  | False | False | False | False | ... | False   | True    | False | True  | False | False |
| 1 | 0.500000 | 0.433962 | 0.175799 | 0.641221 | 0.500000 | True  | False | False | False | True  | ... | False   | False   | False | False | False | False |
| 2 | 0.854167 | 0.481132 | 0.109589 | 0.412214 | 0.419355 | True  | False | False | False | False | ... | False   | False   | False | False | False | False |
| 3 | 0.666667 | 0.509434 | 0.175799 | 0.687023 | 0.000000 | True  | False | False | False | False | ... | False   | True    | True  | False | False | False |
| 4 | 0.687500 | 0.415094 | 0.383562 | 0.267176 | 0.306452 | False | False | False | False | True  | ... | True    | False   | False | False | True  | False |

5 rows × 23 columns

#### Step 7: Final Dataset Check

```
df_encoded.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1025 entries, 0 to 1024
Data columns (total 23 columns):
#   Column          Non-Null Count  Dtype
---  -
0   age             1025 non-null   float64
1   trestbps        1025 non-null   float64
2   chol            1025 non-null   float64
3   thalach         1025 non-null   float64
4   oldpeak         1025 non-null   float64
5   sex_1           1025 non-null   bool
6   cp_1            1025 non-null   bool
7   cp_2            1025 non-null   bool
8   cp_3            1025 non-null   bool
9   fbs_1           1025 non-null   bool
10  restecg_1       1025 non-null   bool
11  restecg_2       1025 non-null   bool
12  exang_1         1025 non-null   bool
13  slope_1         1025 non-null   bool
14  slope_2         1025 non-null   bool
15  ca_1            1025 non-null   bool
16  ca_2            1025 non-null   bool
17  ca_3            1025 non-null   bool
18  ca_4            1025 non-null   bool
19  thal_1          1025 non-null   bool
20  thal_2          1025 non-null   bool
21  thal_3          1025 non-null   bool
22  target_1        1025 non-null   bool
dtypes: bool(18), float64(5)
memory usage: 58.2 KB
```

